

CLOUD COMPUTING



SESSION 1

- Introduction to cloud
- What computing paradigms are there?
- Characteristics and benefits
- Understanding Cloud Vendors (AWS/Azure/GCP)
- Definition
- Characteristics
- Components



Why Cloud Computing?

Hi Sam, I'm about to start a Company.
Can you list down the resources I will need to setup on-premise infrastructure?

Desktops, Servers, Softwares,
Network, Tech Team, Space,
Cooling, Power etc.,

On-
Premise

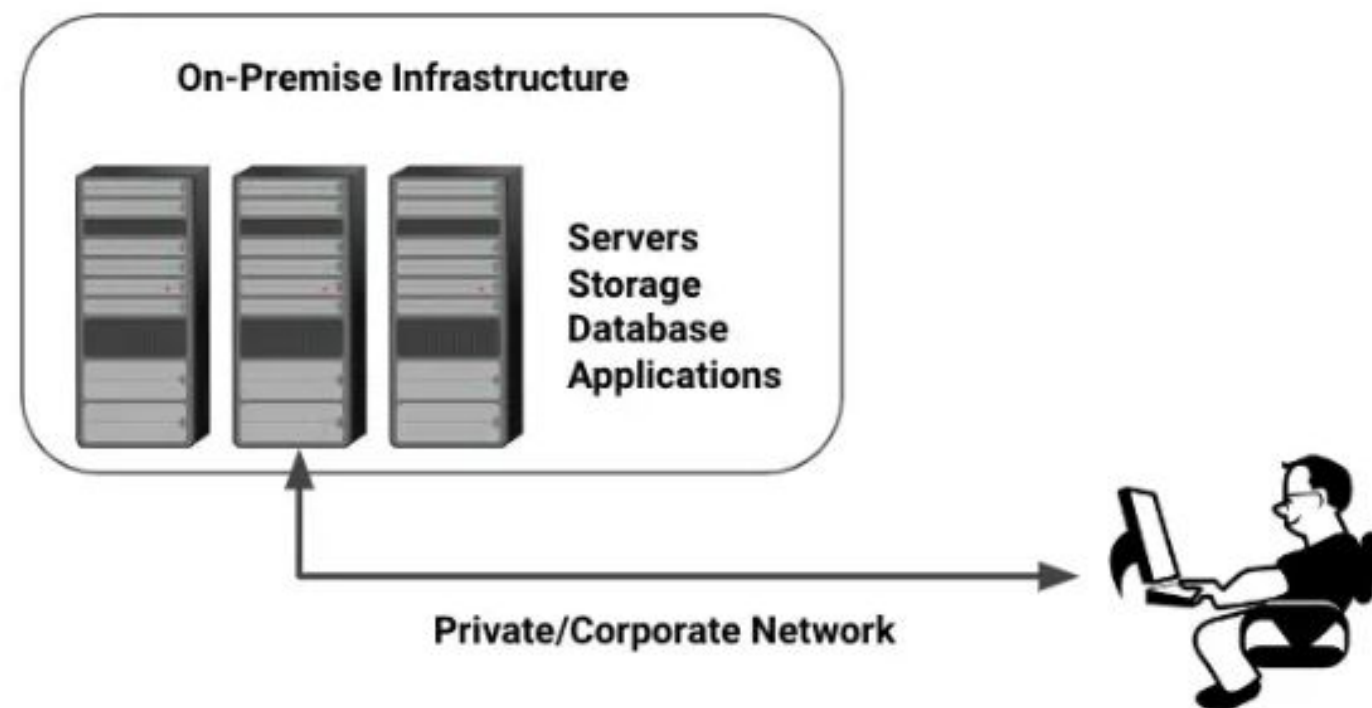
Cloud
Computing



WHY CLOUD COMPUTING

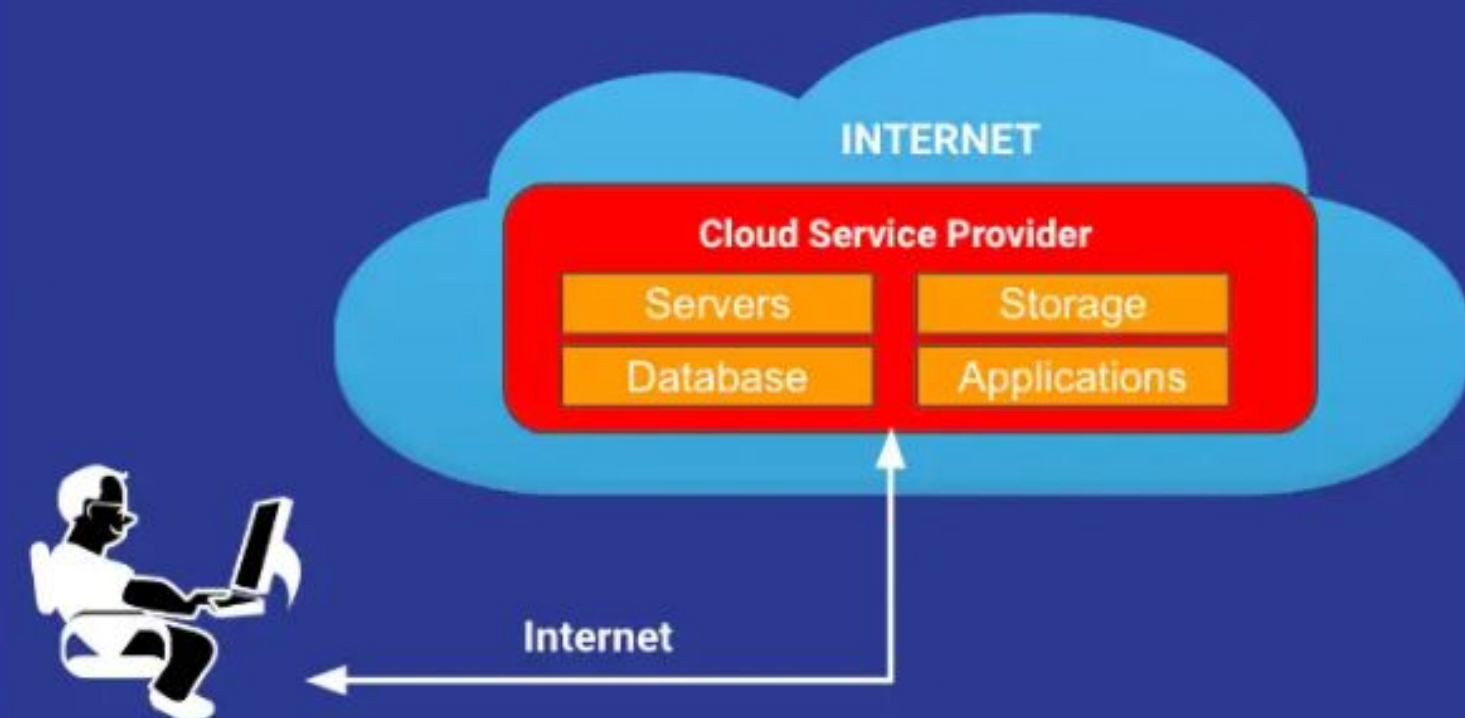
In One Glance

Traditional IT Infra



A traditional IT infrastructure is made up of the usual hardware and software components: **data centers, servers, storage, databases, networking hardware and enterprise applications**. IT Ops team will manage and update it when require.

Cloud Computing



A cloud computing service provides on-demand access to computer system resources, such as computing power, without requiring the user to _____ actively manage them.

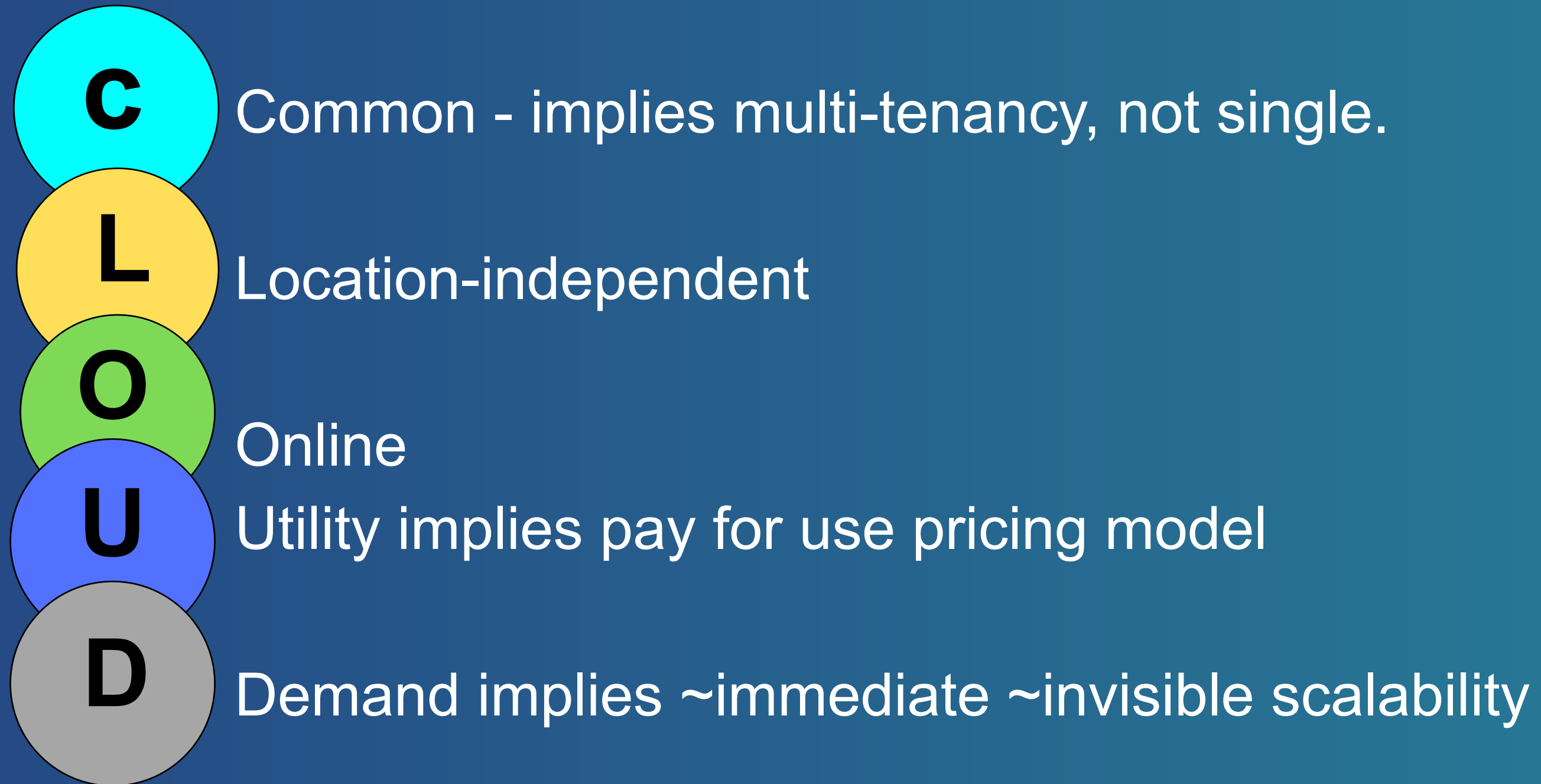


WHAT IS CLOUD COMPUTING

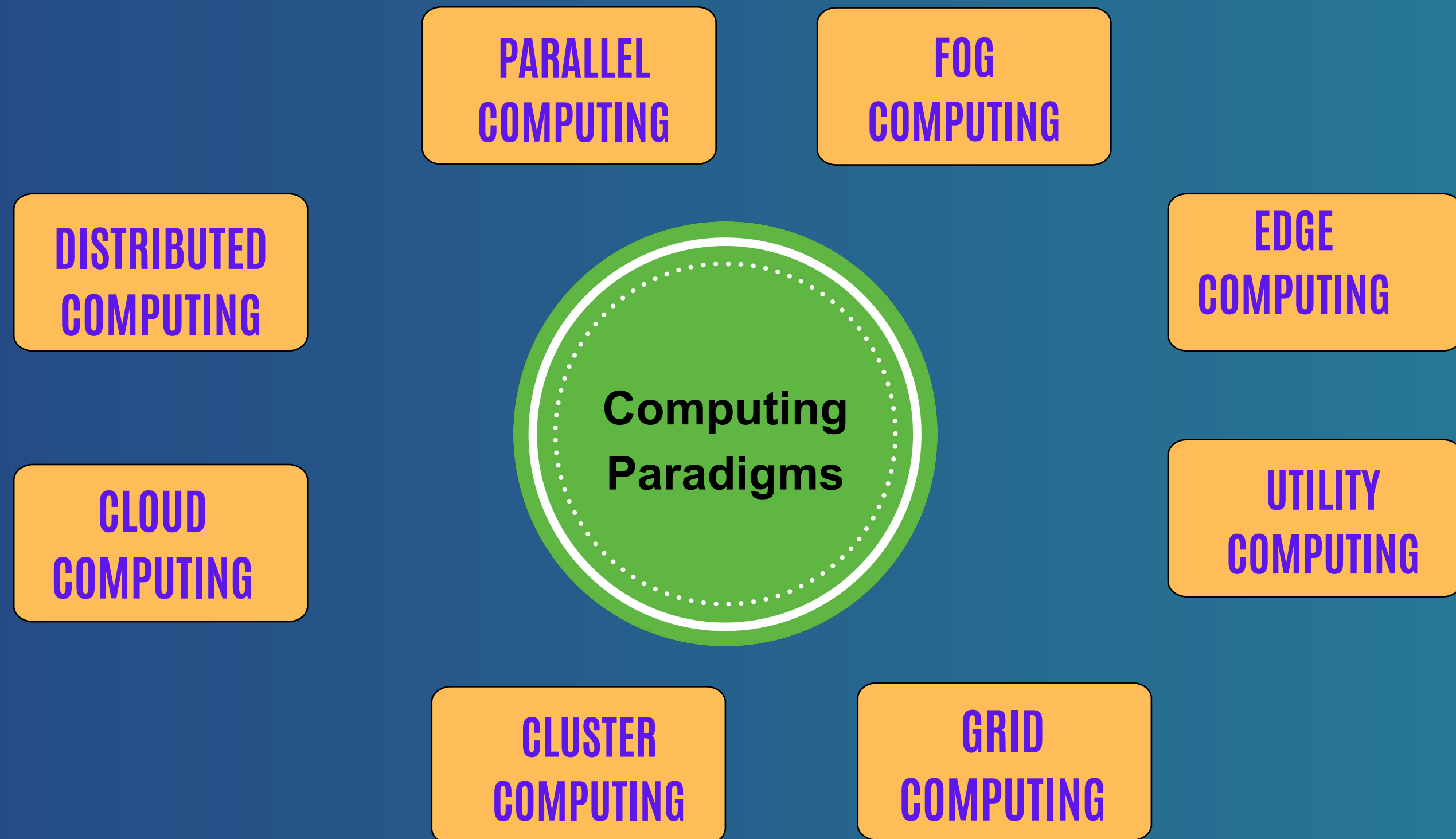
"Cloud Computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g. networks, servers, storage, applications and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction."



The Term “Cloud”?



WHAT COMPUTING PARADIGMS ARE THERE?



WHAT COMPUTING PARADIGMS ARE THERE?

- **Distributed Computing** : Multiple computers work together over a network.
- **Parallel Computing** : Tasks are split and processed simultaneously.
- **Cluster Computing** : Connected computers act as a single system.
- **Grid Computing** : Combines resources from different locations.
- **Utility Computing** : Pay-as-you-go computing services.
- **Edge Computing** : Processes data near the data source.
- **Fog Computing** : Extends cloud services closer to devices.
- **Cloud Computing** : Delivers IT resources over the internet.

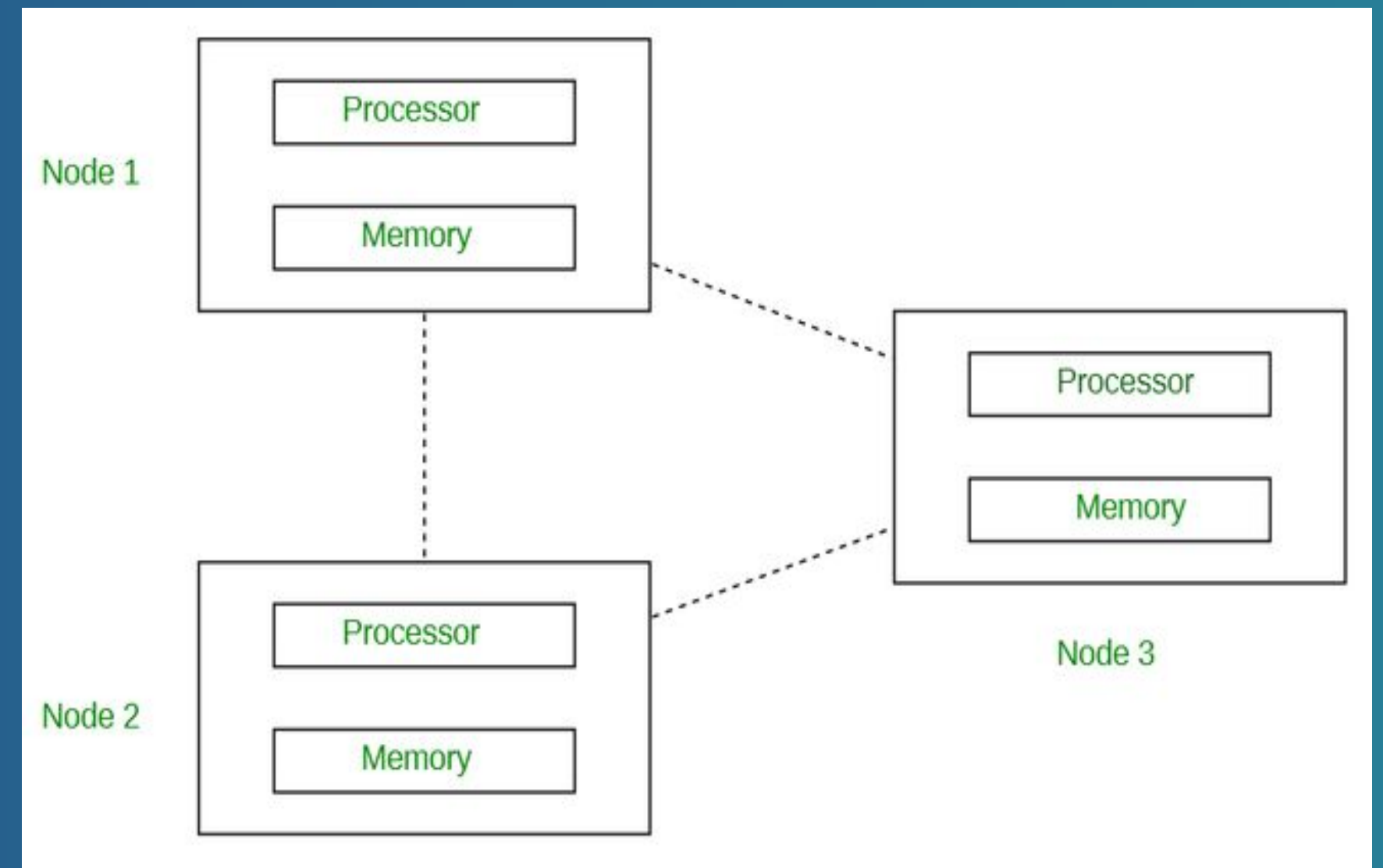


DISTRIBUTED COMPUTING

A computing model where multiple computers connected over a network work together on a single problem.

The problem is divided into sub-problems, and each system solves a part independently.

- Improve performance & speed
- Provide fault tolerance
- Enable scalability for large tasks

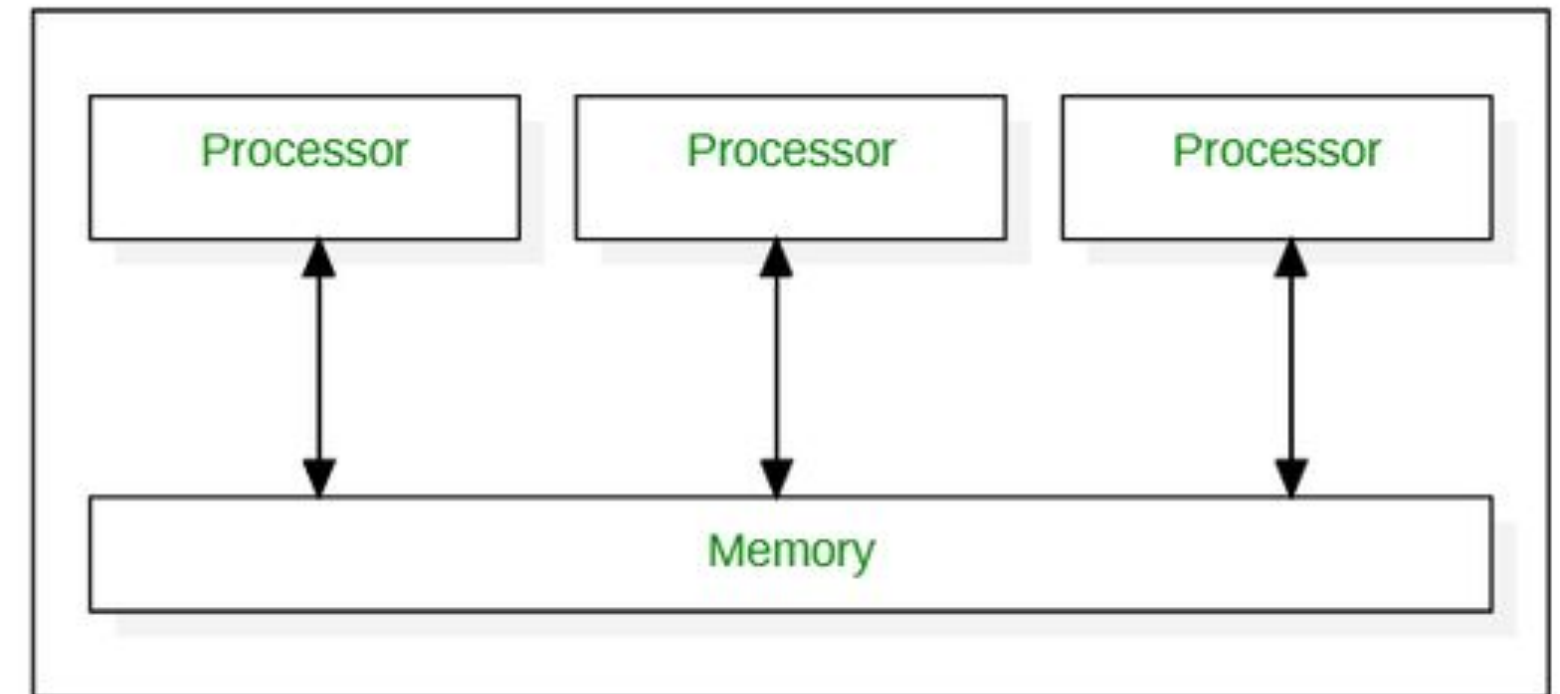


PARALLEL COMPUTING

Parallel Computing is a model where multiple processors work simultaneously to solve a problem by executing instructions concurrently.

A problem is divided into sub-problems, which are further split into instructions that run in parallel on different processors.

- Save time
- Achieve concurrency
- Maximize CPU utilization

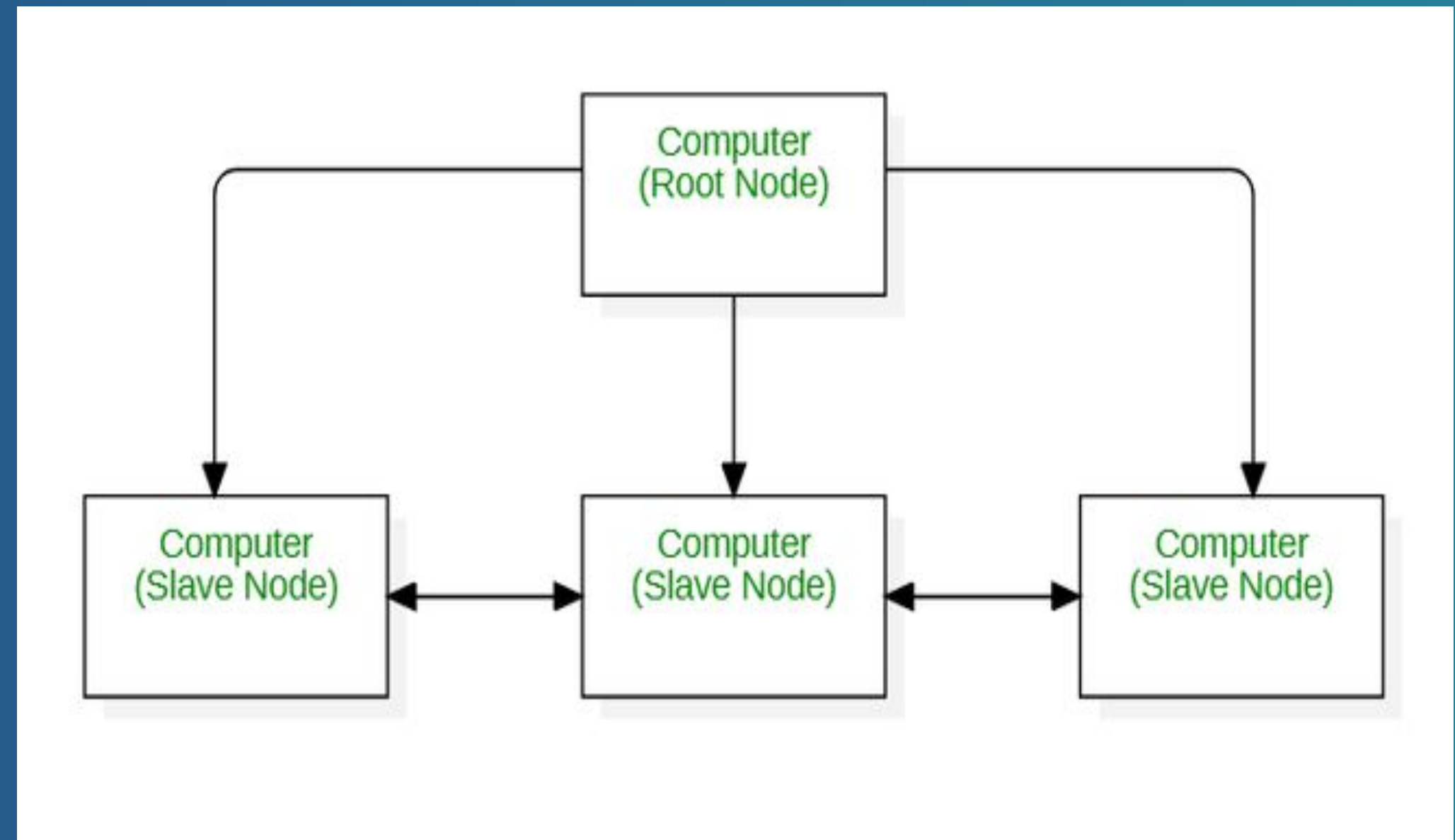


CLUSTER COMPUTING

Cluster Computing is a computing model where two or more independent computers (called nodes) work together as a single system to complete tasks.

These nodes are interconnected and collaborate to enhance computing power.

- Improve performance
- Enable scalability
- Ensure simplicity and reliability

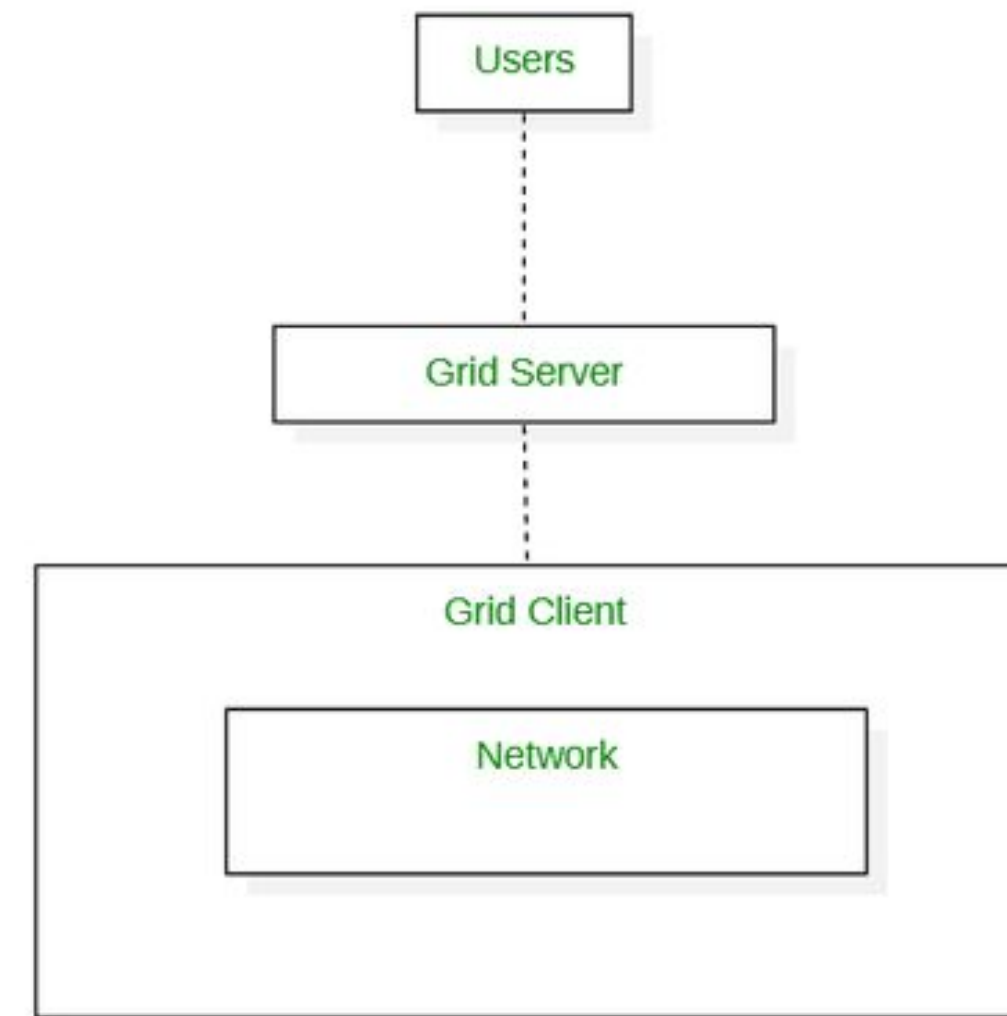


GRID COMPUTING

Grid Computing is a model where a network of geographically distributed computers work together to perform complex tasks that are too large for a single system.

These systems form a virtual supercomputer, sharing resources like CPU, memory, and data.

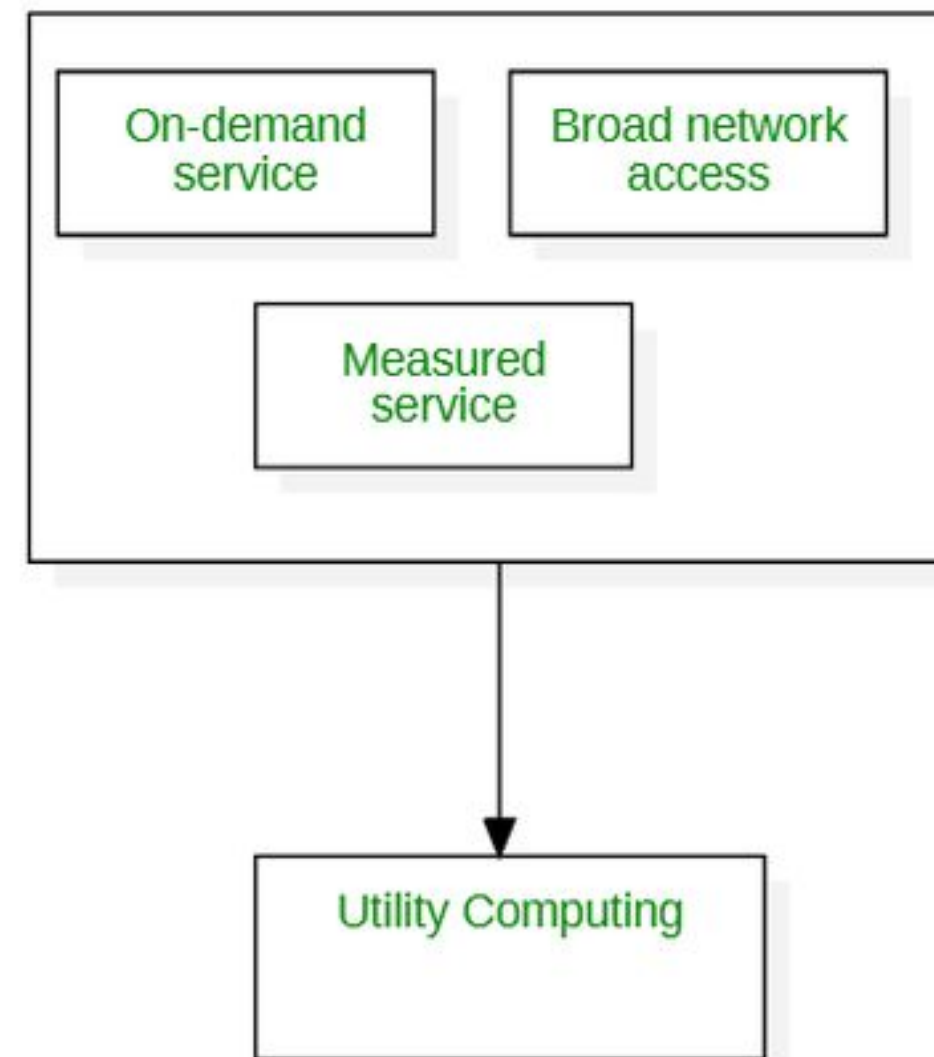
- Solve high-computation problems
- Reduce processing time
- Boost productivity



UTILITY COMPUTING

Utility Computing is a model in which a service provider rents IT resources— hardware, software, storage, or compute power—on-demand and bills customers only for what they use, not a flat rate.

- Maximize resource usage
- Improve cost-efficiency by paying only for what is used

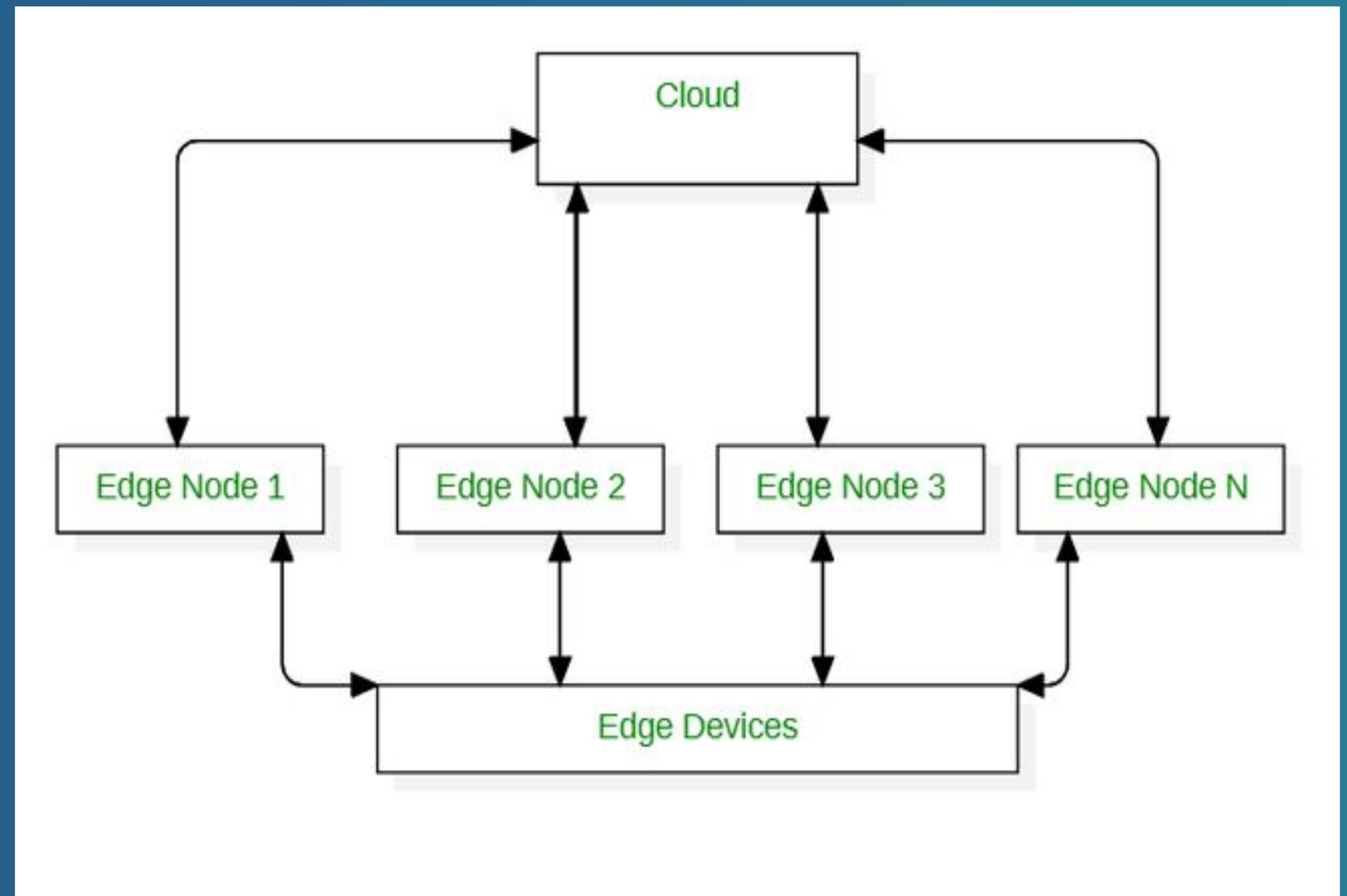


EDGE COMPUTING

Edge Computing is a model where data processing and computation are performed closer to the data source (e.g., user devices, IoT sensors), rather than relying entirely on centralized cloud servers.

This reduces the need for long-distance communication between the client and the server.

- Reduce latency
- Improve real-time responsiveness
- Enable faster decision-making by computing at the network's edge

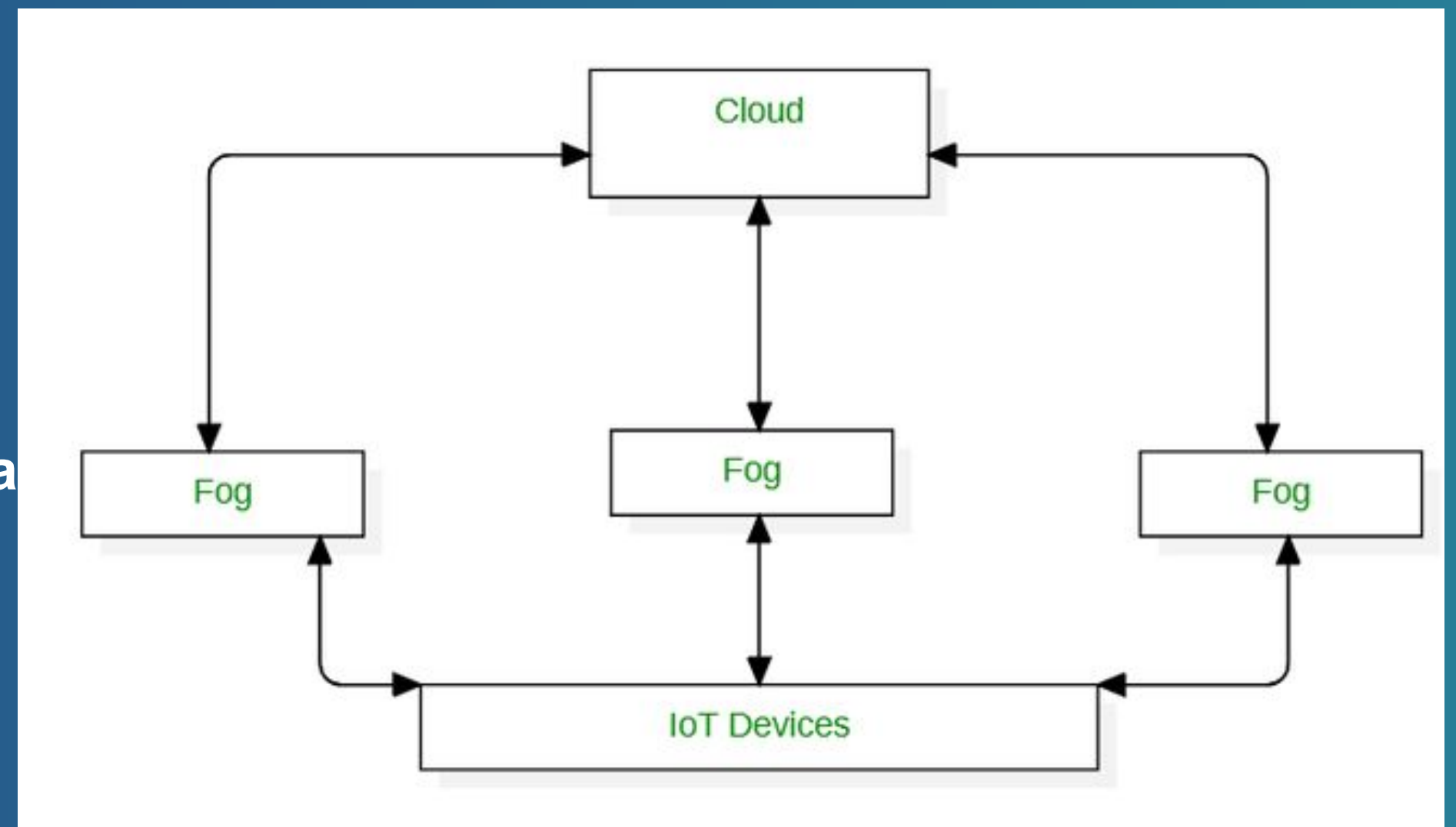


FOG COMPUTING

Fog Computing is a distributed computing model that sits between the cloud and data-producing devices (such as sensors and IoT devices), enabling computation, storage, and networking closer to the source.

This approach, also called "fogging," helps in reducing latency and optimizing data processing.

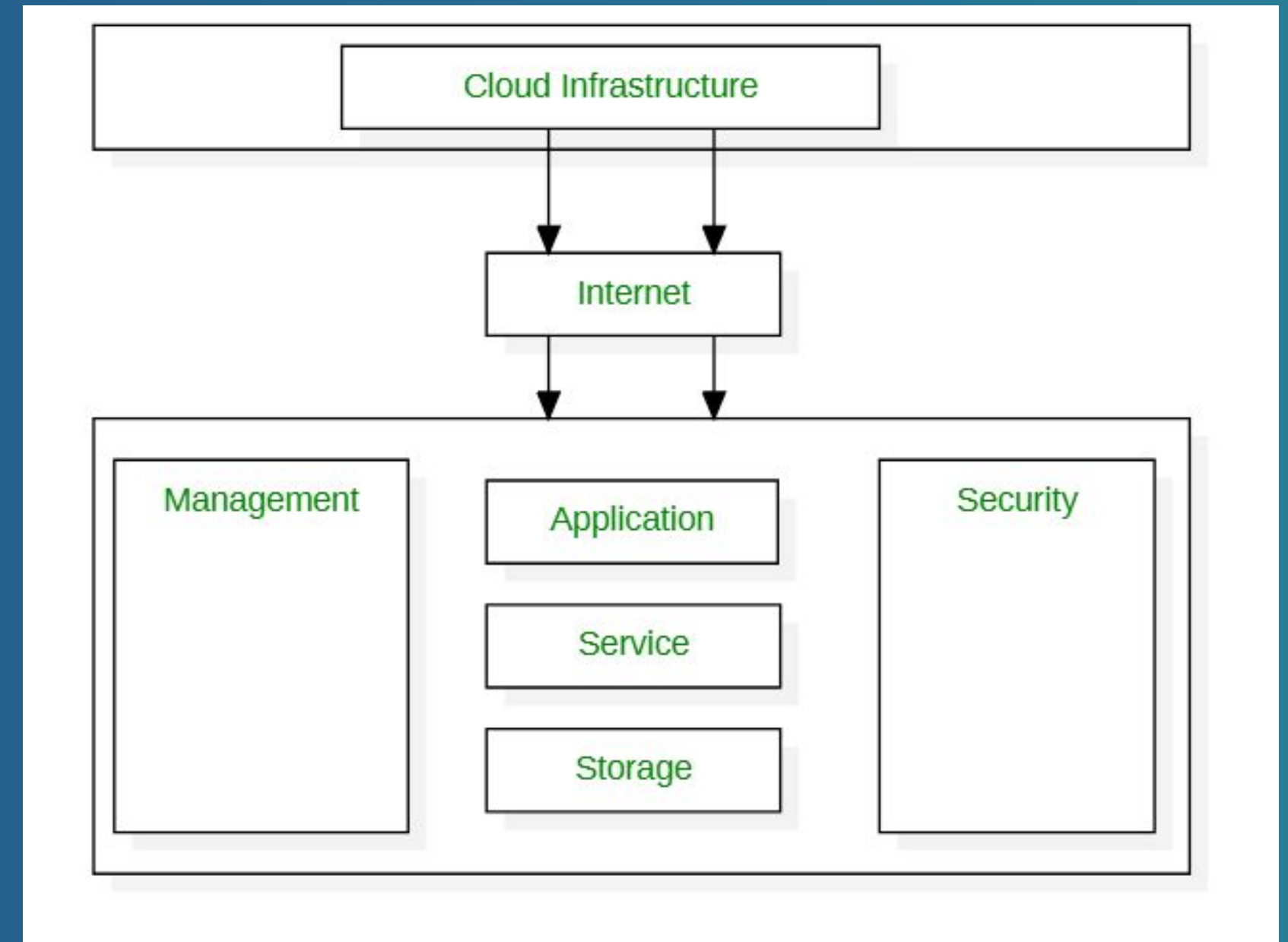
- Improve network efficiency
- Enhance performance by processing data closer to the source
- Reduce latency and bandwidth usage



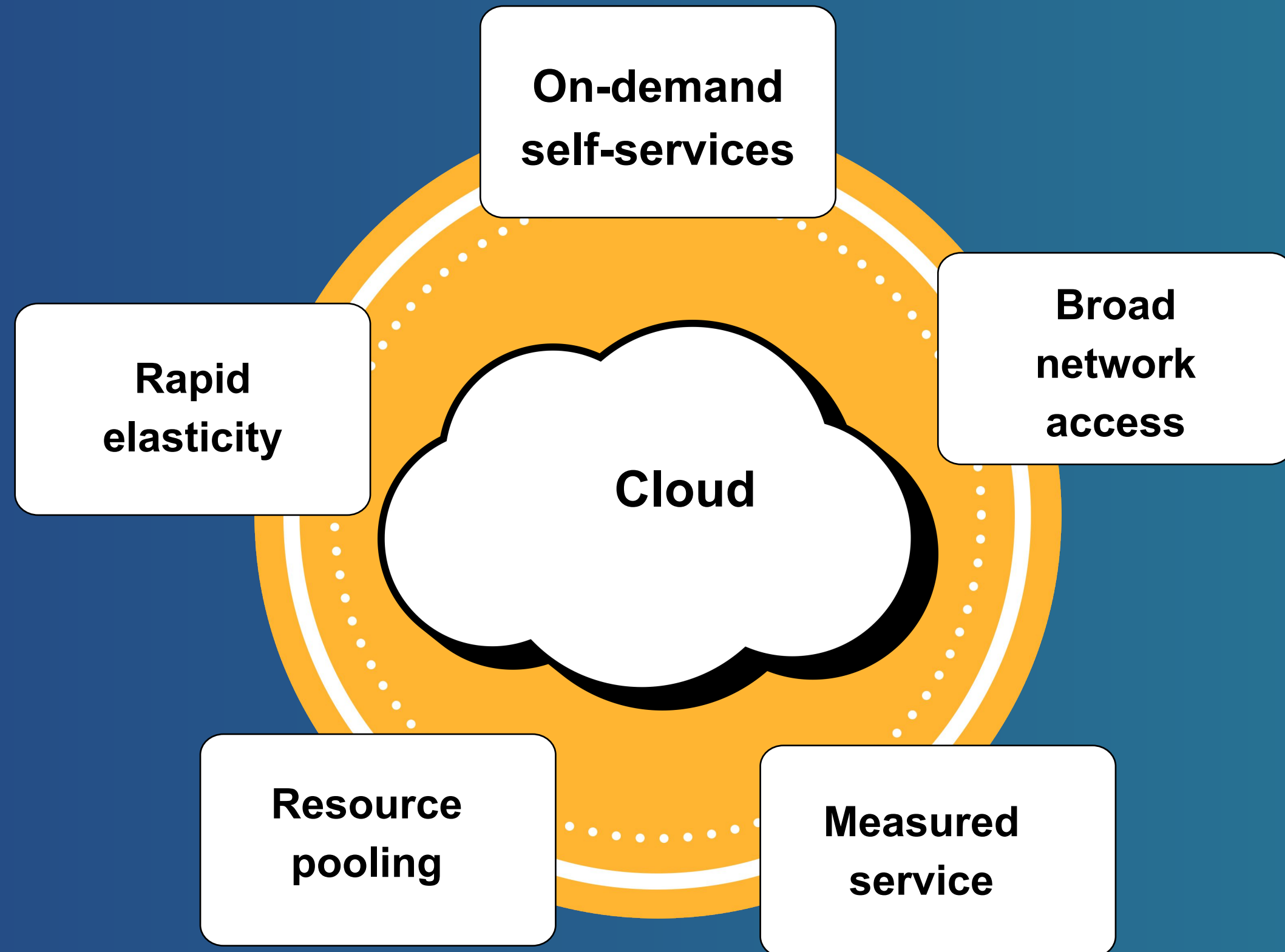
CLOUD COMPUTING

Cloud Computing is the delivery of on-demand computing services (like storage, processing power, and software) over the internet, on a pay-as-you-go basis, using remote servers rather than local infrastructure.

- **Scalable:** Resources can be scaled up or down based on need
- **Pay-as-you-go:** Customers only pay for the resources they use
- **Distributed:** Services are hosted in multiple data centers globally
- **Network-based:** Services are delivered through the internet



CHARACTERISTICS CLOUD COMPUTING



CHARACTERISTICS CLOUD COMPUTING

On-Demand Self-Service :

Users can provision, monitor, and manage computing resources automatically without human interaction.

Broad Network Access :

Services are available over standard networks and can be accessed via various devices (laptops, mobiles, tablets, etc.).

Rapid Elasticity :

Resources can scale up or down quickly based on demand. Services are provisioned instantly and released when no longer needed.

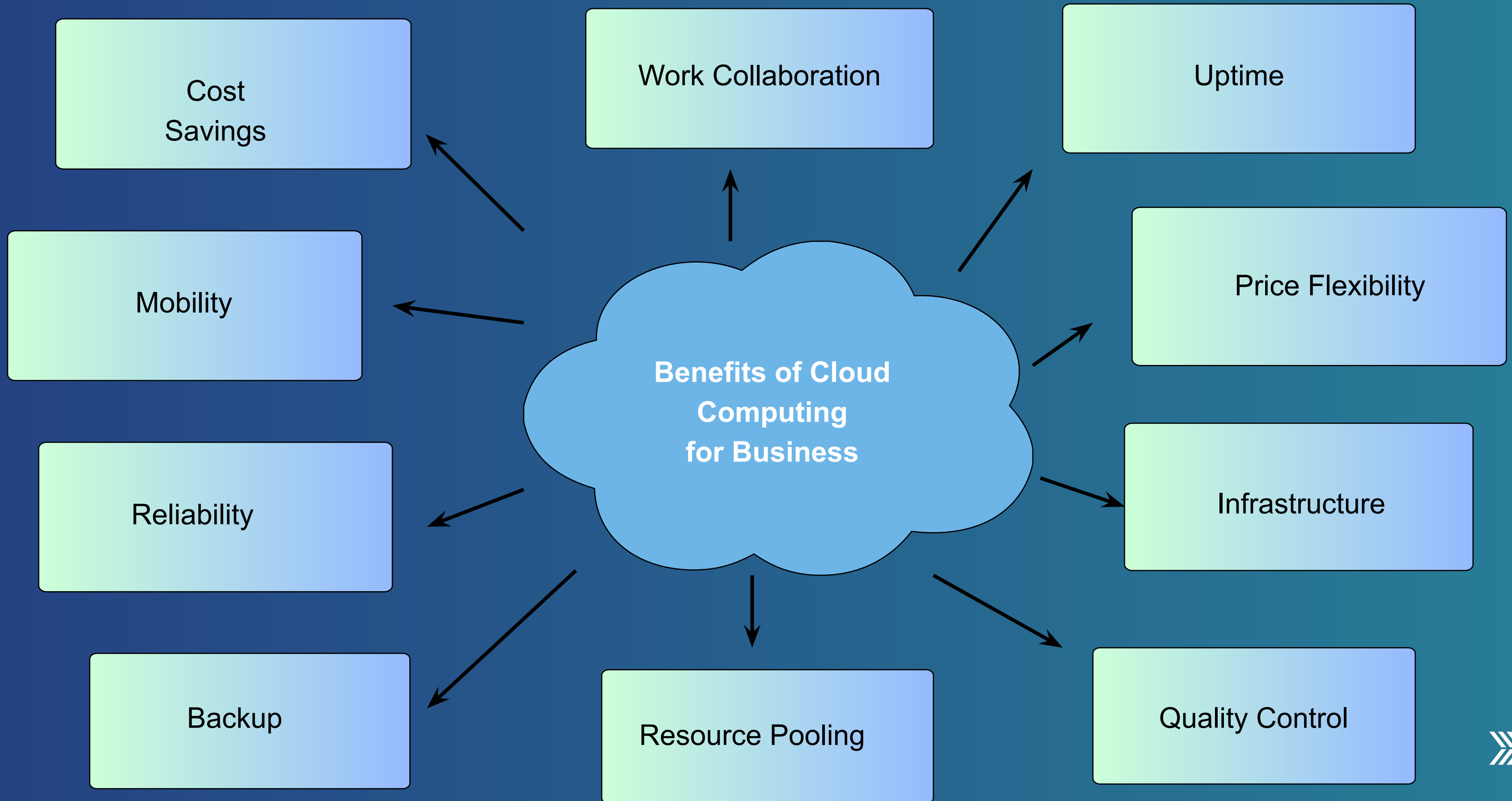
Resource Pooling :

Shared resources (servers, storage, networks, etc.) serve multiple users dynamically, offering flexibility and efficiency.

Measured Service :

Resource usage is monitored and recorded, enabling transparent billing and optimized resource allocation.

BENEFITS OF CLOUD COMPUTING



UNDERSTANDING CLOUD VENDORS

Top 10 Cloud Providers



Source: <https://allcode.com/cloud-providers/>

AMAZON WEB SERVICES (AWS)

Overview:

- AWS is a comprehensive and widely adopted cloud platform by Amazon.
- Offers IaaS, PaaS, and SaaS solutions.

Types of Cloud Services:

- IaaS: EC2 (virtual servers), VPC, S3 (storage), EBS.
- PaaS: AWS Lambda, Elastic Beanstalk, RDS.
- SaaS: AWS WorkMail, AWS Chime.

Key Features:

- Highly scalable and flexible.
- Global infrastructure with data centers in multiple regions.
- Pay-as-you-go pricing model.

Use Cases:

- Web hosting, big data processing, machine learning, app development.

DROPBOX

Overview:

- Dropbox is a SaaS platform that focuses on cloud storage and collaboration.
- Allows users to store, sync, and share files easily.

Type of Cloud Service:

- SaaS (Software as a Service)

Key Features:

- Cloud-based file storage and synchronization.
- Team collaboration tools (comments, shared folders).
- File recovery and version history.

Use Cases:

- Personal file backup.
- Business document sharing and collaboration.
- Remote access to files from any device.

GOOGLE CLOUD PLATFORM (GCP)

Overview:

- GCP is Google's suite of cloud computing services.
- Provides IaaS, PaaS, and SaaS.

Types of Cloud Services:

- IaaS: Compute Engine, Cloud Storage, Virtual Networks.
- PaaS: App Engine, Cloud Functions, Cloud SQL.
- SaaS: Google Workspace (Docs, Sheets, Gmail, Meet).

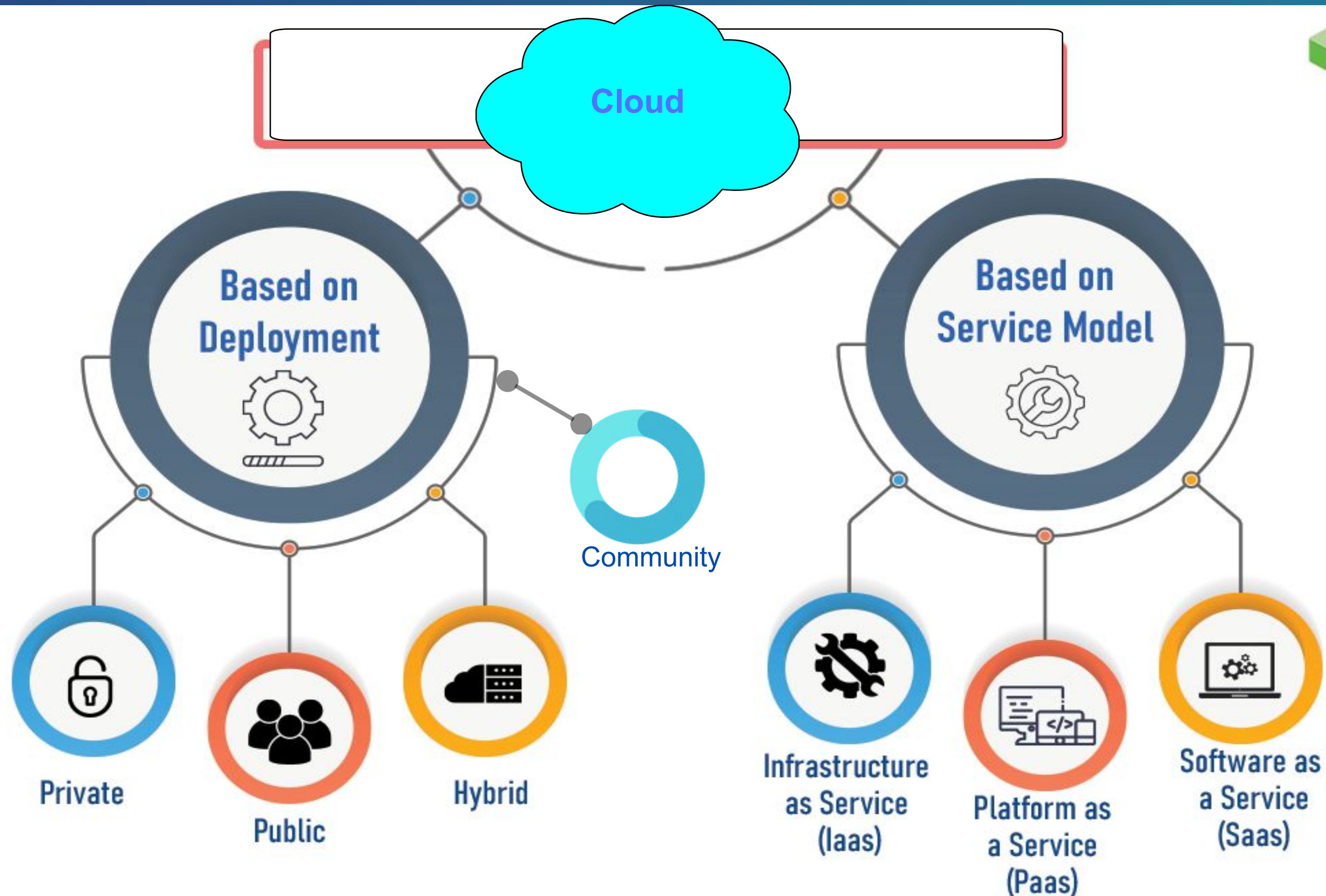
Key Features:

- Strong AI/ML tools (e.g., Vertex AI).
- High-performance global network.
- Cost-effective and environmentally friendly.

Use Cases:

- AI/ML development, data analytics, app deployment, collaboration tools.

TYPES OF CLOUD COMPUTING



CLOUD DEPLOYMENT MODEL

Public Cloud

Typically have massive amounts of available space, which translates into easy scalability. Recommended for software development and collaborative projects.

Private Cloud

Usually reside behind a firewall and are utilized by a single organization. Recommended for businesses with very tight regulatory requirements.

Deployment Models

Hybrid Cloud

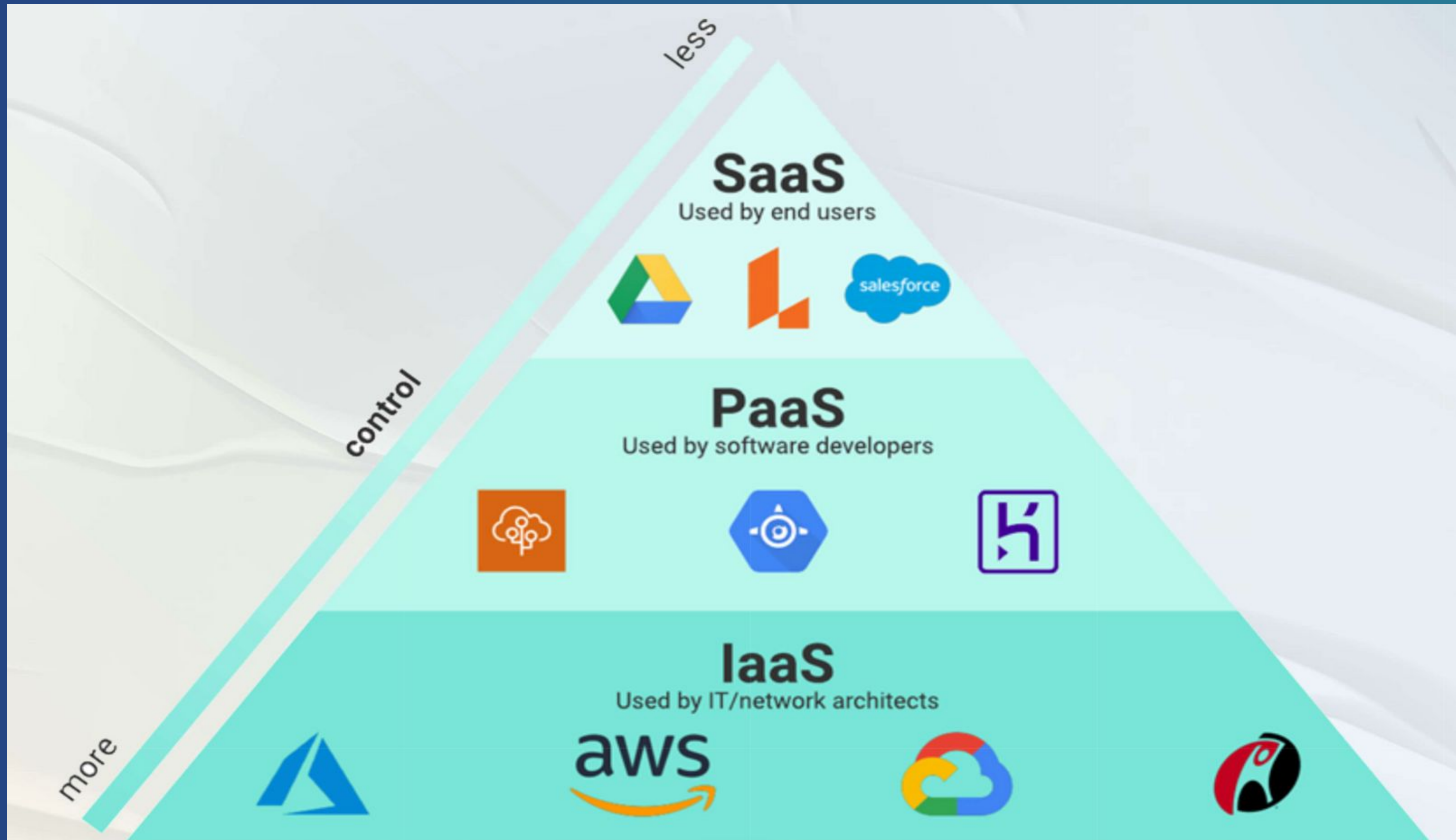
Combine public clouds with private clouds to allow the two platforms to interact seamlessly. Recommended for businesses balancing big data analytics with strict data privacy regulations.

Community Cloud

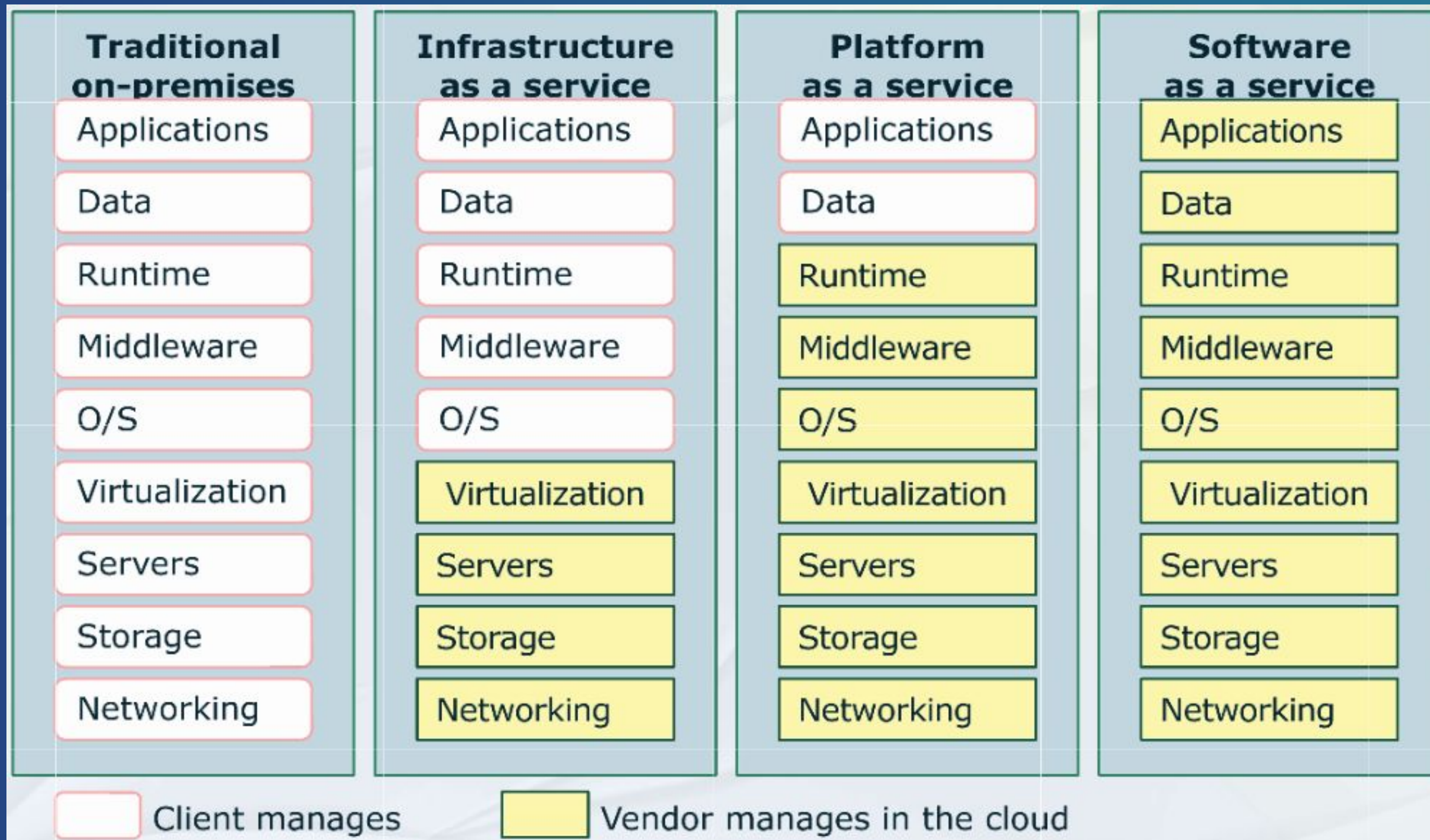
A collaborative, multi-tenant platform used by several distinct organizations to share the same applications. Users are typically operating within the same industry or field.



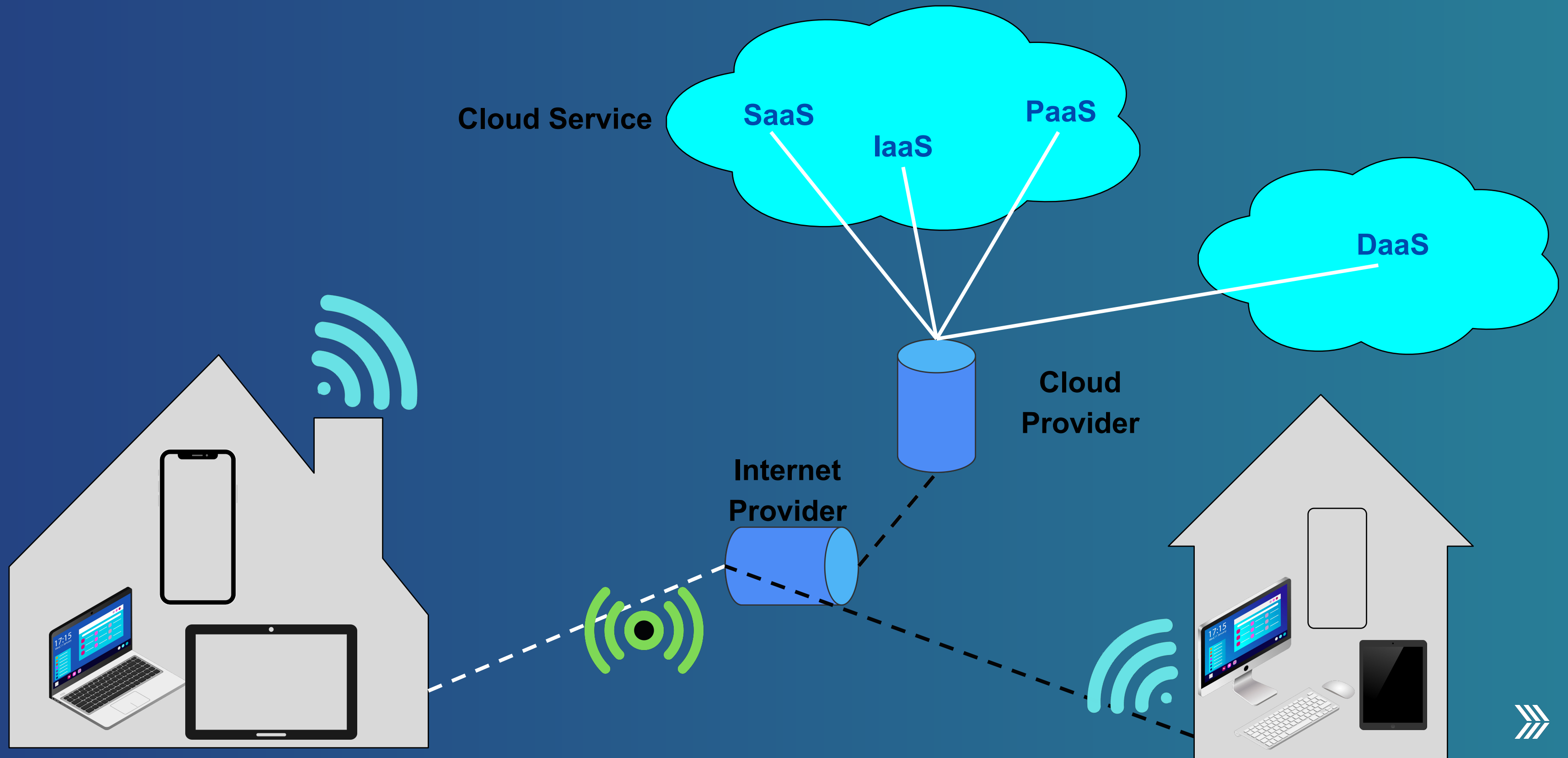
CLOUD SERVICE MODELS



SPLIT OF RESPONSIBILITY



CLOUD NETWORK TOPOLOGY



CLOUD COMPONENTS

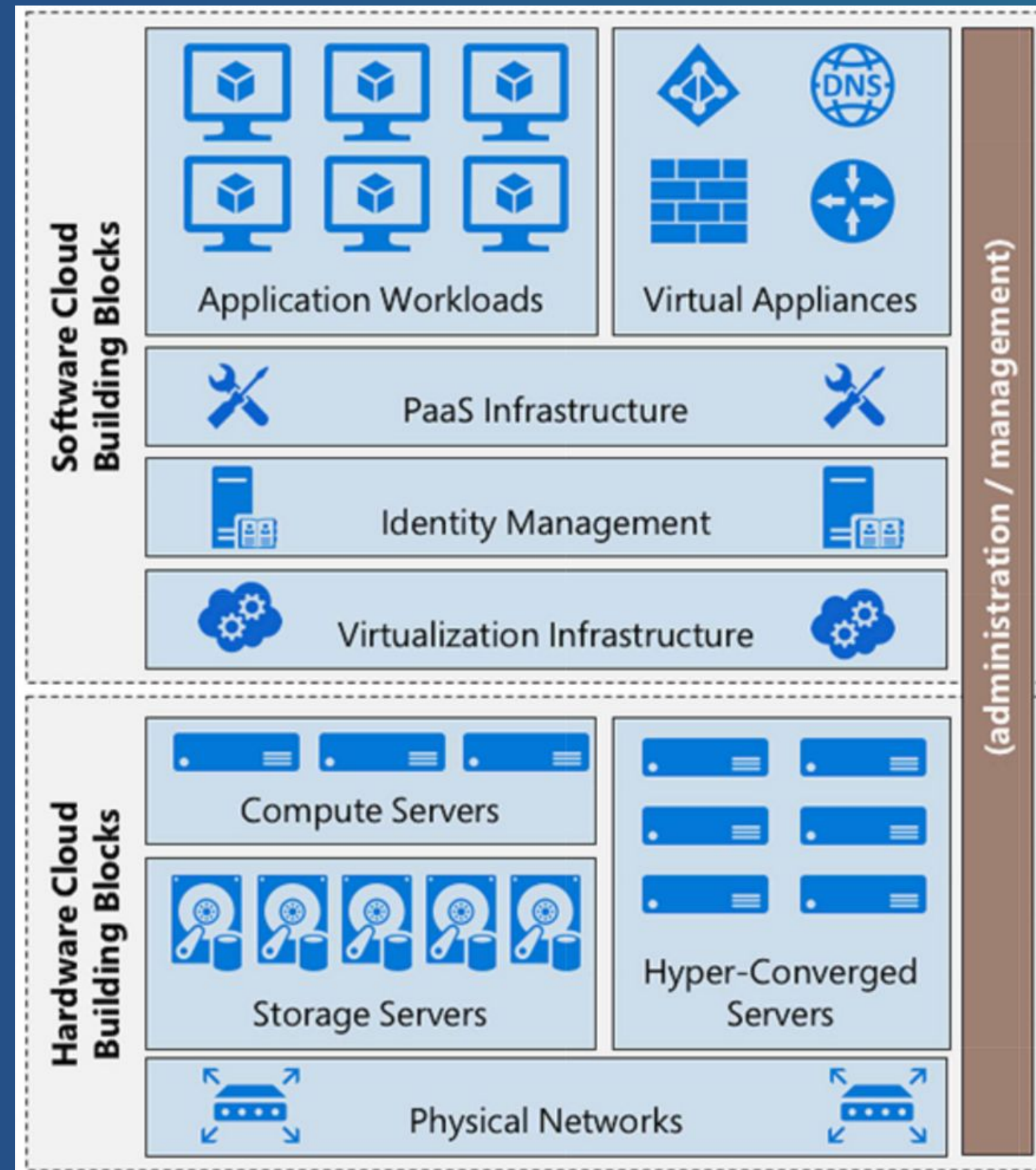
Application

Middleware

Hypervisor

**Hardware
Resources**

CLOUD LAYER ARCHITECHURE

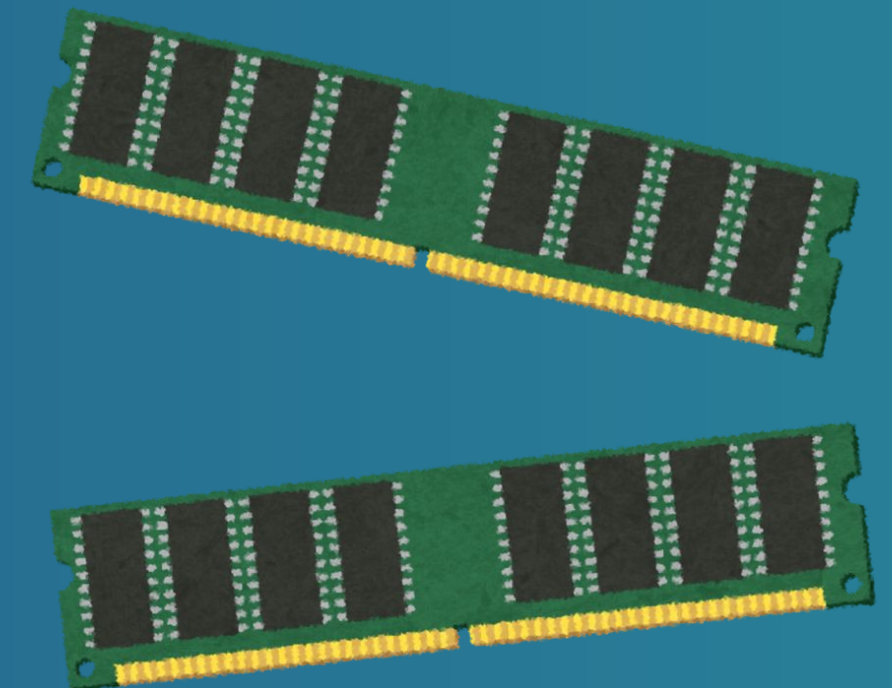
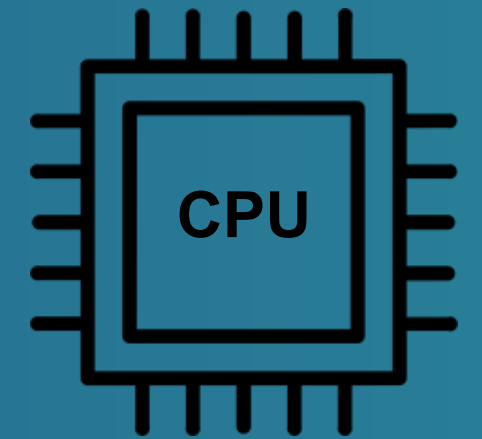


HARDWARE RESOURCES

Physical infrastructure that powers the cloud.

Components:

- Servers, CPUs, memory, storage drives, network equipment, power systems.
- Foundation for virtualized resources and services.
- Hosted in data centers and accessed remotely.
- High availability and performance.
- Redundancy and fault tolerance.
- Supports virtualization and scalability

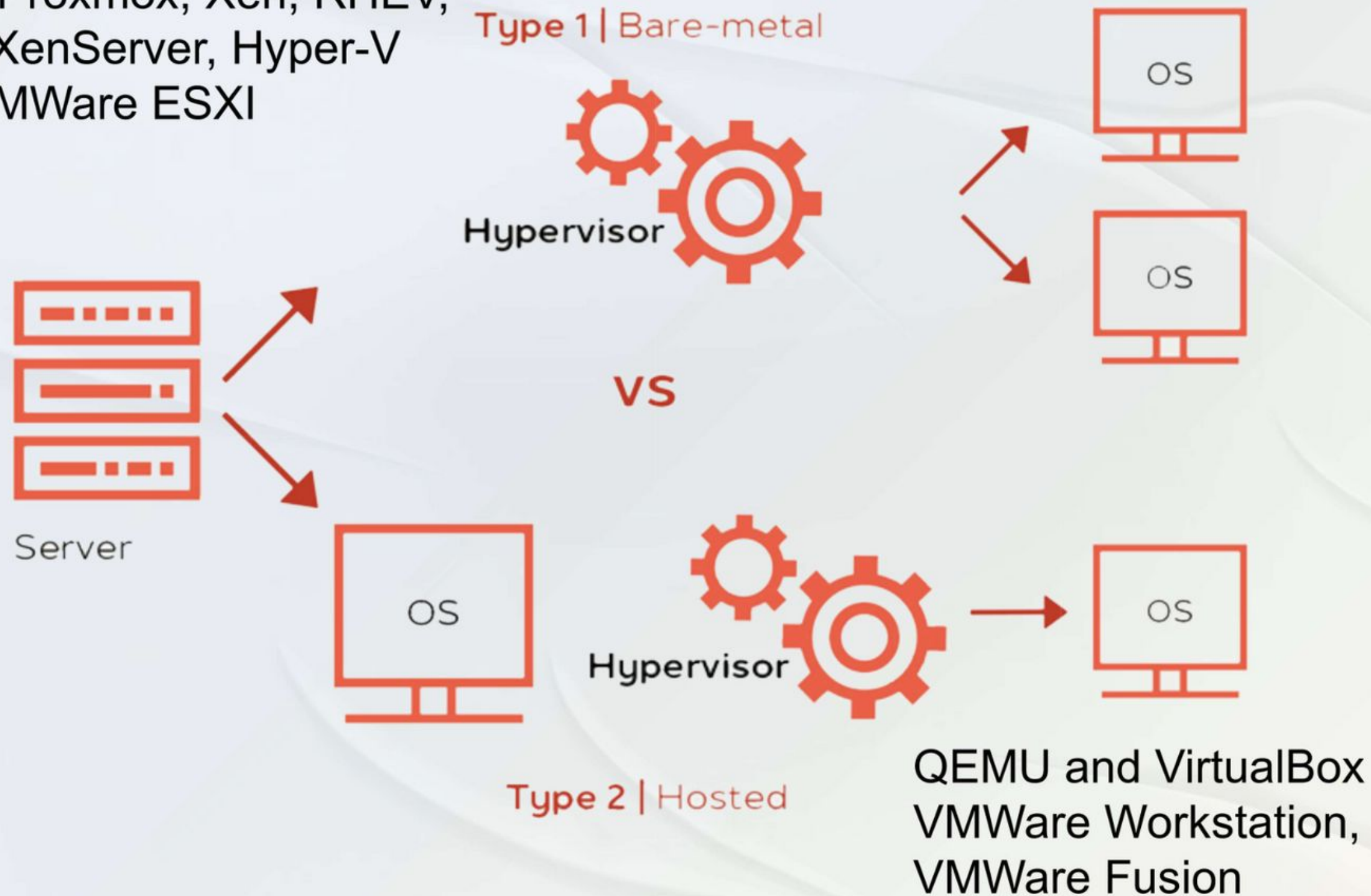


HYPERVISORS

- The most important component of cloud
- The term "hypervisor" is considered the supervisor of "supervisors".
- It is a thin layer software
- A hypervisor, also known as a **Virtual Machine Monitor (VMM)**, is a **layer of virtualization software** that allows the creation and running of several virtual machines within a single server, as well as different operating systems.
- The server where the hypervisor runs one or more virtual machines is commonly known as a "**host machine**", while each individual VM is commonly known as a "**guest machine**".
- Dynamically Partitioning and Sharing the physical resources such as **CPU, storage, memory and I/O devices**.

HYPERVISORS

KVM, Proxmox, Xen, RHEV,
Citrix XenServer, Hyper-V
and VMWare ESXI



CLOUD MIDDLEWARE

A software platform that sits between an application/device and another application/device.

**Openstack
Apache Cloudstack
OpenNebula
Amazon EC2
Google Cloud Engine
Microsoft Azure**

Management and Monitoring

**Resource
scheduler**

Data management

**Identity/Security management
User**

Interfaces and portale

Billing and metering

Key Features

APPLICATION

Applications are the software programs delivered to end-users via the cloud.

Role in Cloud:

- **Users interact with cloud services through applications (e.g., web apps, mobile apps).**

Examples:

- **Google Docs, Salesforce, Dropbox, Microsoft 365.**



LIFE-CYCLE OF A CLOUD COMPUTING SOLUTION

